Speech-Eval

Contrastive Speech Delivery Analytics with Temporal Flaw Grounding

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1. Problem and approach

Speech-Eval locates delivery differences in pace, loudness and pauses, with word/time playback and explanations measured against readers or your earlier take.

Frozen A: precision 0.125, recall 0.400, F1 0.190.
Frozen B: precision 0.018, recall 0.390, F1 0.035. Genuine v1: A 4/13 hits, 77 FP; B 3/13 hits, 60 FP.
A clean alarms: 1.341/min; dev guard failed (1.777 versus <=0.5/min).
Release: 1,103 indexed recordings, 36.3 h; repeated scientific hashes match.
Limits: low precision; genuine labels are recorder-confirmed, not independent; zero rater responses.

Content quality is out of scope. Controlled edits supply operation labels, not independent judgments. A uses same-text train readers, B the earlier take, C-lite sentence norms. Key-moment heuristics do not affect scores.

2. Contrastive dataset construction

Sources: 15 Gettysburg and 13 O Captain LibriVox readings, plus one JFK Rice and one Reagan Challenger recording. USA public-domain evidence is recorded in the release manifests and dataset card; automatic evidence checks are not human legal review. Raw hashes remain immutable.

Local CTC transcript checks used 300 calibration controls (fake acceptance ≤2.0%); this in-sample proxy is not listening or guaranteed accuracy. Unsettled words/dependent evidence remain masked; number expansion preserves indices. Acoustic audit.

MFA/WhisperX were compared on 37 words / 74 edges after 23 unclear-word exclusions. Boundaries were placed visually from waveform and spectrogram by an AI assistant (Claude), without listening; no human spot-check was done. Independence from model templates was asserted, not independently established. Pooled median differences: MFA 52.0 ms, WhisperX 72.0 ms; MFA is primary. Both passed independent CPU rerun identity. These are differences from a limited visual reference, not established human accuracy. Bake-off.

Eligible sources receive seven effects at five edit strengths plus mixed variants, with seeded placement. One clipped O Captain source is audit-only. The release has 1,103 indexed variants / 36.3 h: 30 originals and 1,073 edited recordings. Raw downloads/duplicate rates are not counted again. Train/dev/test have 18 / 4 / 7 speakers. Measured counts.

Type	Controlled operation	Increasing edit strength
rushing / dragging	Pitch-preserving stretch	Raise / lower speed factor
monotone	PSOLA around median pitch	Retain less pitch range
too_quiet	Flat/trailing gain	More attenuation
awkward_pause	Phrase-internal room tone	Longer inserted gap
missing_pause	Shorten existing gap	Lower retained fraction
mumbling	Zero-phase low-pass plus gain	Lower cutoff/gain

Parameters are in configs/flaws.yaml. Time maps transport source indices into the flawed timeline; labels bind type, edit-strength severity and interval. Real MFA is run separately for detection. Oracle alignments serve label/QA only; edit boundaries are not guaranteed perceptual boundaries. Genuine recorder-confirmed takes are evaluated separately below, without threshold tuning.

3. Features and speaker normalisation

Praat pitch/intensity/HNR and spectral/MFCC tracks feed word, short-window and sentence aggregates. Speaker-relative transformations reduce voice/microphone offsets, not all differences:

$st(t) = 12 \log_2[f_0(t) / \text{median } f_0]$; $Lrel(t) = L(t) - \text{median } L$.

Medians use voiced frames inside aligned words. Rate (syllables/second) is not speaker-normalised. Clarity is a high-band energy ratio, not intelligibility. Comparison rate/pitch-variation/pause features are log-transformed; loudness and clarity remain dB. Uncertain-word masks exclude dependent aggregates. Feature normalisation does not silently edit stored loudness.

4. Detection, grounding and explanations

Clean train readers give word/feature median mu and robust spread $s = 1.4826 \text{ median}(|x - mu|)$. Pooled train spread and shrinkage $n0 = 5$ give:

$\sigma^2_{i,f} = (n\, s^2_{i,f} + n_0\, \sigma^2_{pool,f}) / (n + n_0)$; $z_{j,f} = (x_{j,f} - \mu_{i,f}) / \sigma_{i,f}$.

Whole-word mapping excludes unmatched/partial correspondences. Runs within sentences become regions; pauses are boundary events. Directional feature rules and contradictory-pair suppression are explicit. Frozen selected k is **3.50 for every type**, not the draft default 2.0. Dev severity cut points are [2.0, 2.75, 3.5, 4.059, 6591.0]; matched dev Spearman is 0.16. The unusual final cut point/weak correlation are retained, not repaired with test or user takes. Calibration.

Templates quote measured participant/reference deviations and flagged words. Real presentation-layer output, unchanged evidence from Mode A test analysis 86964ad39ead, gettysburg__lv_sa__awkward_pause_s3_01:

> 75.99–77.58 s, "it can": you paused 1.59 s between "it" and "can", where good readers do not pause (1.59 s longer, more than 10 SD above). It breaks the sentence.

The matching label is 76.035-77.535 s: a hit under the frozen pause-centre rule. Full-precision reference/z remain in JSON. Extreme z reflects near-zero spread, not perceptual severity. This existing illustration was not used for v2 tuning; measured explanations are **not causal proof**.

B uses the own earlier take; JFK/Reagan test originals are within-speaker references, not population training/calibration. C checks sentence rate, pitch range, final loudness drop and non-final end pause; mumbling/awkward pause are unsupported. Number/end guidelines never alter scores.

5. Scoring and validation

For first-to-last-word duration D, duration penalty is $\text{sum}(v \times d/D \times 10)$; pause penalty is $\text{sum}(0.4 \times v)$. Clip 10 - penalty to [0,10], decimal half-up round, then weight rounded categories: pace/pitch 0.25 each, pauses 0.20, volume/clarity 0.15 each. These are configurable design choices, not an external standard. Severity-3 rushing over 10.0% gives pace 7.0. Validation.

Per-type monotonicity (seven complete series each):

Type	Median severity-score Spearman	Non-increasing %	Constant series
awkward_pause	-0.87	85.7	3
dragging	-0.71	71.4	2
missing_pause	-0.71	100.0	4
monotone	-0.71	28.6	0
mumbling	-0.89	71.4	0
rushing	-0.71	71.4	1
too_quiet	-0.71	85.7	3

Constant ranks have undefined Spearman; non-increasing alone is not sensitivity. Human agreement was not measured: no independent raters were available; no AI or self-ratings were used. The blind pack asks overall delivery 1–5, flaw types and time. Scripts compute system/human and defined pairwise Spearman plus exact flaw-set agreement with denominators; no responses or CIs are fabricated.

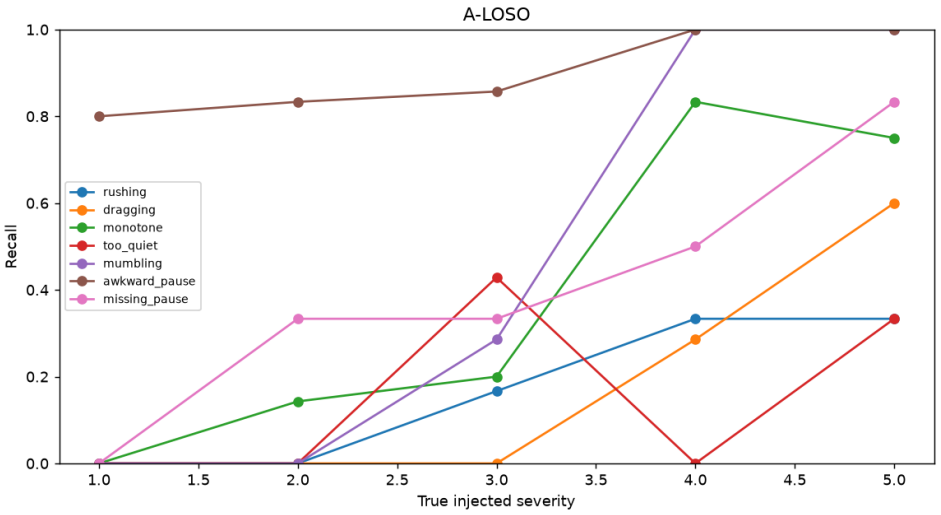
Frozen host two-run identity: 10/10. Later fresh-clone CPU Docker reproductions matched all 563 scientific outputs twice, including cross-environment checks. Scope and SHA manifests.

6. Results across settings

Thresholds used dev only. Original final test commit: aaf1832fe24a193d4c90f9e7a2ff9537b753335c. Later test runs are **reproductions**, not tuning. Same-type one-to-one region IoU ≥0.3; pause centre distance ≤0.25 s. Frozen results.

Setting	Files	Precision	Recall	F1 [95% CI]	Clean alarms/min
A-LOSO	190	0.125	0.400	0.190 [0.156, 0.223]	1.341
B-same-speaker	76	0.018	0.390	0.035 [0.026, 0.045]	0.000

A highest recall: awkward_pause 0.903; lowest: rushing 0.167. Matched overall onset/offset differences are 26.0 / 40.5 ms (A), 31.0 / 35.5 ms (B), conditional on matches, not all truth regions. Dev clean-alarm guard **failed**: 1.777 versus ≤0.5. Learned detector was not trained. No pooled severity-1/2 or severity-4/5 recall is inferred from the chart without a reported numeric summary.



A-LOSO recall versus injected severity

6.1 Secondary detector v2

post-hoc v2; test split was seen with v1, so results are optimistic. V1 above remains the primary frozen result. V2 was selected from exactly 30 DEV configurations and frozen at 95c8f334faf4c134fcbf5b4fa31fda4635f336ea before one marker-protected test evaluation. No detector/configuration changes followed. Within-recording sentence contrast, practical effect minima and region constraints raised DEV F1 from 0.150 to 0.219 with 0.410 clean alarms/min, passing the declared promotion gate. The selected configuration SHA is 3496a9c3....

Setting / version	Files	Precision	Recall	F1 [95% CI]	Clean/min
A frozen v1	190	0.125	0.400	0.190 [0.156, 0.223]	1.341
A post-hoc v2	190	0.145	0.288	0.193 [0.154, 0.232]	0.671
B frozen v1	76	0.018	0.390	0.035 [0.026, 0.045]	0.000
B post-hoc v2	76	0.042	0.207	0.070 [0.043, 0.099]	0.000

Rows, matcher, exclusions and denominators are identical; zero uncertain truth exclusions on these test rows. Per-type/severity results include boundaries, alarms and severity Spearman. Strength-cohort precision/F1 uses shared clean controls and homogeneous truth strengths, excluding mixed strengths; full-set severity recall includes them. Full precision/hashes: v2 JSON. For A, v2 halves clean alarms (1.341 to 0.671 per min) but lowers recall (0.400 to 0.288). F1 is effectively unchanged (0.190 to 0.193); v1 stays primary. The test clean-alarm guard still fails. B precision/F1 rise but remain low, with recall falling. V2 is a precision/recall trade-off, not established general improvement; uniform recording-wide problems can be missed by within-recording contrast. Human takes/ratings were not used for tuning.

6.2 Mode B precision diagnosis (already completed, DEV only)

Frozen v1 compared 152 DEV Gettysburg/O Captain variants with their own originals: 70 hits, 94 misses, 554 FP; precision 0.112, recall 0.427. Four self-controls had zero flags. FP positions use the nearest edit of any type; wrong-type inside flags still count as FP. This is separate from the published B test cohort.

FP type	Before	After	Inside	Overlap
awkward_pause	284	232	19	11
rushing	0	0	5	1
dragging	0	0	1	0
missing_pause	0	0	1	0

Median absolute pause FP difference was 0.008 s. After rushing/dragging/pause edits, real-MFA start residuals versus mapped source were median 0.003 s / p95 0.009 s, versus 0.287 / 2.204 s without mapping. Detection uses real MFA, not oracle time maps. Outside-edit HQ samples differed by at most 0.596 PCM16 LSB after recorded global gain; 37 files had logged anti-clipping gain. Map duration error was <0.000499 s. Two rounded insertion-knot oracle edge outliers remain disclosed; they are not detector inputs. Tiny pause spread amplifies alignment jitter; no demonstrated live B timing/processing bug was found. No fix, retuning or new test run was made. Detailed existing DEV audit.

7. No-reference comparison

C uses **different eligibility and sentence/type matching**. Its paired comparison on 175 files stays separate from Phase6:

Paired setting	Precision	Recall	Clean alarms/min
Same-text / own-original reference	0.106	0.324	0.584
C-lite norms	0.020	0.179	4.159

C too-quiet truth is trailing-only; unsupported single effects are omitted and mixed files score supported types only. Lower aggregate recall/more clean alarms describe controlled edits, not humans. Mode C.

7.1 Genuine recorder-confirmed takes

Eight consented takes completed real MFA: A for all, B02/03 versus own01, C for both spontaneous take04s (unknown flaws, excluded from scoring). Labels are recorder-confirmed, not independent; word anchors were fixed before results. Transcripts apply ASR structural edits but keep canonical substitutions and mumbling words; verbatim fidelity is not independently verified. No tuning.

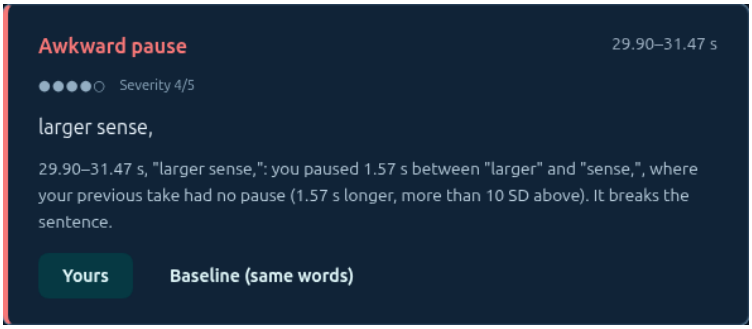
Version / mode	Labels	Hits / misses / FP	FP/min
Frozen v1 / A	13	4 / 9 / 77	9.625
Frozen v1 / B	13	3 / 10 / 60	11.236
Post-hoc v2 / A	13	4 / 9 / 58	7.250
Post-hoc v2 / B	13	3 / 10 / 27	5.056

Primary matching is same-type one-to-one IoU >=0.3, including pause events. A pools six scored takes (8.00 min); B pools four current takes (5.34 min), not their references. Take01 flags: Gettysburg/Reagan 9/14 (v1), 7/11 (v2). Historical pause-centre <=0.25 s, secondary unit overlap, MFA spans, transcripts and dashboard links: Phase10. V1 stays primary; post-hoc v2; test split was seen with v1, so results are optimistic. Limited hits/high false alarms do not establish independent perceptual accuracy.

Takes finish sooner than every same-text reference: Gettysburg 80.99-86.81 s versus 102.88-190.95 s; Reagan 76.42-80.29 s versus a 90.17 s excerpt. Total aligned gap seconds are lower than nine Gettysburg readers and Reagan; only Gettysburg take03 is below the tenth reader too. This is not every sentence or pause. Mode A's 53 unlabelled missing pauses (of 77 v1 false alarms) are measured reader differences, counted as false alarms because unlabelled, never hits.

8. Dashboard

Both screenshots show genuine Gettysburg take02 in frozen v1 Mode B versus own take01: the declared larger|sense hit at 29.90-31.47 s, with waveform, scores, measured explanation and same-words playback. Genuine browser provenance.



Selected flaw region: genuine take02 vs take01, declared pause hit



Dashboard results: genuine Gettysburg take02 Mode B, waveform and score card (excerpt; full screenshot)

9. Limitations and future work

Controlled edits are not real co-occurring flaws, and neutral reading is not ideal oratory. Few texts/speakers and English-only proxies limit generalisation. The unheard visual reference and unclear-word exclusions introduce uncertainty and selection bias; model-to-visual differences do not establish annotation accuracy. C/key-moment heuristics and rubric scores lack human validation. Low precision, the failed dev guard, constant/non-monotonic series and fixture miss remain limits. Future work needs more consented recordings and independent labels/ratings; detector changes must be labelled post-hoc, not presented as frozen evaluation.

10. Reproducibility and attributions

make reproduce-quick downloads the main archive by Drive ID with pinned gdown, verifies SHA256 and runs one text/three readers. Browser fallback uses DATA_ARCHIVE; dataset card has exact files/hashes/commands. The separate human-takes archive is owner-confirmed; remote re-download/check was waived. make reproduce reproduces frozen Phase6/7/8 numbers in pinned CPU Docker, not historical ASR decisions or human/video work. Quick: 11.829 / 11.954 seconds; full: 225.204 / 223.142 after import/build. Hashes: data/release/reproduction_hashes.json. The disclosed legacy-fixture exemption is a post-hoc release adapter, not a detector change; other frozen inputs/code are verified.

Code: MIT; labels, annotations, docs and my recordings: CC BY 4.0, with consent. Source audio remains public domain per source.

Measured retained reproduction image + one ZIP + unpacked payload: 32,170,019,620 bytes (32.170 GB); 16.804 + 7.587 + 7.779 GB respectively. This excludes build/extraction scratch and host caches; the retained image predates the gdown metadata layer, so it is a snapshot, not a clean-build peak. Exact disk accounting.

Audio credit: LibriVox volunteers and JFK/Reagan institutional collections, with source-specific USA public-domain evidence.

Pending items

Recorded/published video URL.

Clean-build peak unmeasured. Remaining actions: NEXT.